Chapter-VIII

MAGNETISM IN SOLIDS

Magnetism was observed as early as 800 BC in a naturally occurring material called load stone which was used for navigation purposes. In the modern concept, all materials, viz., metals, semiconductors and insulators, are said to exhibit magnetism, though of different nature. The present chapter reviews the magnetic terminology in brief and describes the various types of magnetism in detail.

8.1 MAGNETIC TERMINOLOGY

When a solid is placed in a magnetic field, it gets magnetised. The magnetic moment per unit volume developed inside a solid is called *magnetization* and is denoted by **M**. There is another important parameter called the *magnetic susceptibility*, χ, which is a measure of the quality of the magnetic material and is defined as the magnetization produced per unit applied magnetic field, i.e.,

$$\chi = \mathbf{M}/\mathbf{H} \tag{8.1}$$

where **H** is the strength of the applied magnetic field, also referred to as the *magnetic field intensity*. As the vectors **M** and **H** can, in general, have different directions, χ is a tensor. However, in isotropic media, **M** and **H** point in the same direction and χ is a scalar quantity. In the SI system, both **M** and **H** are measured in amperes per metre (Am^{-1}) and hence χ is a dimensionless quantity. If **M** refers to a gram molecule of a substance, the susceptibility is termed the *molar susceptibility* and is designated by χ_m. The magnitude and sign of susceptibility vary with the type of magnetism as will be discussed later.

The *magnetic induction* or *magnetic flux density* **B** produced inside the medium as a consequence of the applied magnetic field **H** is given by

$$\mathbf{B} = \mu_0 (\mathbf{H} + \mathbf{M}) \tag{8.2}$$

where μ_0 is the *permeability* of the free space or vacuum and is equal to $4\pi \times 10^{-7}$ Henry per metre (Hm^{-1}). The quantity **B** is measured in Weber per square metre (Wbm^{-2}) or Tesla (T). Using Eq. (8.1) in (8.2), we obtain

$$\mathbf{B} = \mu_0 (1 + \chi) \mathbf{H} \tag{8.3}$$

or $$\mathbf{B} = \mu \mathbf{H} \tag{8.4}$$

where μ is called the *absolute permeability* of the medium. Like χ, μ is also, in general, a tensor; for the isotropic medium, however, it represents a scalar quantity having dimensions same as that of μ_0. It is more convenient to introduce a dimensionless parameter μ_r which is called the *relative permeability* of the medium and is given by

$$\mu = \mu_0 \mu_r \tag{8.5}$$

Thus, Eq. (8.4) becomes

$$\mathbf{B} = \mu_0 \mu_r \mathbf{H} \tag{8.6}$$

From Eqs. (8.3) and (8.6), we get

$$\mu_r = 1 + \chi \tag{8.7}$$

For free space, i.e., in the absence of any material medium, $\mathbf{M} = 0$, $\chi = 0$, $\mu = \mu_0$ and $\mu_r = 1$, and from the above relations, we obtain

$$\mathbf{B} = \mu_0 \mathbf{H} \tag{8.8}$$

8.2 TYPES OF MAGNETISM

According to the modern theories, the magnetism in solids arises due to orbital and spin motions of electrons as well as spins of the nuclei. The motion of electrons is equivalent to an electric current which produces the magnetic effects. The major contribution comes from the spin of unpaired valence electrons which produces permanent electronic magnetic moments. A number of such magnetic moments may align themselves in different directions to generate a net non-zero magnetic moment. Thus the nature of magnetization produced depends on the number of unpaired valence electrons present in the atoms of the solid and on the relative orientations of the neighbouring magnetic moments. The magnetism in solids has been classified into the following five categories :

(i) Diamagnetism

(ii) Paramagnetism

(iii) Ferromagnetism

(iv) Antiferromagnetism

(v) Ferrimagnetism

Diamagnetism is a very weak effect and is observed in solids which do not contain any permanent magnetic moments. The existence of a small non-zero magnetic moment in these materials is attributed to the orbital motion of electrons. This magnetic moment is always directed opposite to the applied magnetic field. The other types of magnetisms arise due to the presence of permanent atomic or electronic magnetic moments. *Paramag-*

netism is also a weak effect but, unlike diamagnetism, the magnetic moment is aligned in the direction of the field. *Ferromagnetism* is a very strong effect and arises when the adjacent magnetic moments align themselves in the same direction. In *antiferromagnetism,* the adjacent moments are equal and opposite to each other and hence complete cancellation of moments takes place. *Ferrimagnetism* is similar to antiferromagnetism except that adjacent moments are unequal in magnitude and hence complete cancellation of moments does not take place. The magnetic materials are classified according to the nature of the magnetism present. It may, however, be noted that above a certain critical temperature, the distinction between various types of magnetic materials disappears and all the materials become paramagnetic. The various types of magnetisms have been described in detail in the following sections.

8.3 DIAMAGNETISM

As indicated above, diamagnetism in a material arises due to changes in the atomic orbital states induced by the applied magnetic field. An electron revolving in an orbit constitutes an electric current. When a magnetic flux linked with such an electric circuit is changed, an induced current is set up in such a direction as to oppose the change in flux in accordance with the Lenz's law. The magnetic field of the induced current is opposite to the applied field and produces the diamagnetic effect. The occurrence of diamagnetism is manifested by the very small and negative value of the magnetic susceptibility. Diamagnetism exists in all materials but is usually suppressed due to the presence of stronger effects such as paramagnetism, ferromagnetism, etc. The diamagnetism has been described quantitatively by applying the classical and quantum theories.

8.3.1 Langevin's Classical Theory

Consider an electron revolving around the nucleus in a circular orbit with frequency ω_0. Since a moving electron is equivalent to a current, some magnetic flux is linked with such a current loop. When an external magnetic field is applied, the magnetic flux linked with this current loop tends to change which alters the current in the loop in such a way as to oppose the change in flux. This change in current is manifested by the change in the frequency of revolution of the electron. If ω_0 is the frequency of revolution of the electron in the absence of an applied field and ρ is the radius of the orbit, then we have

$$m\omega_0^2\rho = \frac{Ze^2}{4\pi\varepsilon_0\rho^2} \tag{8.9}$$

where Ze represents the nuclear charge and m is the mass of the electron. It gives the value of ω_0 as

$$\omega_0 = \sqrt{\frac{Ze^2}{4\pi\varepsilon_0 m\rho^3}} \tag{8.10}$$

In the presence of a magnetic induction **B**, the Lorentz force on an electron moving with velocity **v** is given by

$$\mathbf{F} = e(\mathbf{v} \times \mathbf{B}) \tag{8.11}$$

Assuming the magnetic field to be perpendicular to the orbit, the magnitude of this force is

$$F = e\rho\omega B \tag{8.12}$$

If ω is the freqency of electron in the presence of the field, the force equation can be written as

$$m\omega^2\rho = \frac{Ze^2}{4\pi\varepsilon_0\rho^2} - e\rho\omega B$$

$$= m\omega_0^2\rho - e\rho\omega B \qquad \text{[Using Eq. (8.10)]} \tag{8.13}$$

where the minus sign with the Lorentz force represents the negative charge. Thus

$$\omega^2 + \frac{eB}{m}\omega - \omega_0^2 = 0$$

or
$$\omega = \frac{1}{2}\left[-\frac{eB}{m} \pm \sqrt{\left(\frac{eB}{m}\right)^2 + 4\omega_0^2}\right]$$

$$= \pm\sqrt{\omega_0^2 + \left(\frac{eB}{2m}\right)^2} - \frac{eB}{2m}$$

For $\frac{eB}{2m} \ll \omega_0$, we obtain

$$\omega \cong \pm\,\omega_0 - \frac{eB}{2m} \tag{8.14}$$

This shows that the frequency of revolution of an electron changes by a factor

of $\frac{eB}{2m}$ in the presence of the magnetic induction **B**. The ± sign in (8.14) signifies that the electrons with orbital moments parallel to the field are slowed down whereas those with moments antiparallel to the field are speeded up. This result is called the *Larmor theorem*. The change in frequency produces an additional current which can be written as

$$I = \text{charge} \times \text{revolutions per unit time}$$

$$= (-e)\left(\frac{1}{2\pi}\frac{eB}{2m}\right)$$

$$= -\frac{e^2B}{4\pi m} \tag{8.15}$$

The magnetic moment of the current loop containing a single electron is given by the product of the current and the area of the loop, i.e.,

$$\mu_e = -\frac{e^2B}{4m}\rho^2 \tag{8.16}$$

The negative sign signifies that the induced magnetic moment points in a direction opposite to the applied field. If **B** is not perpendicular to the plane of the orbit, then ρ in Eq. (8.16) represents the projection of the radius of the orbit on a plane perpendicular to the magnetic field. If an atom contains Z electrons with their orbits randomly oriented, the total induced magnetic moment becomes

$$\mu_a = -\frac{Ze^2B}{4m}<\rho^2> \tag{8.17}$$

If the field points along z-axis, then

$$<\rho^2> = <x^2> + <y^2>$$

is the mean of the squares of perpendicular distances of electrons from the axis of the field. The mean square distance of electrons from the nucleus is

$$<r^2> = <x^2> + <y^2> + <z^2>$$

For a spherically symmetric charge distribution, we have

$$<x^2> = <y^2> = <z^2>$$

$$<r^2> = \frac{3}{2}<\rho^2> \tag{8.18}$$

Equation (8.17) now gives

$$\mu_a = -\frac{Ze^2B}{6m}\langle r^2\rangle = -\frac{\mu_o Ze^2H}{6m}\langle r^2\rangle \tag{8.19}$$

For a solid consisting of N atoms per unit volume, each containing Z electrons, the expression for the *diamagnetic susceptibility* becomes

$$\chi_{dia} = \frac{M}{H} = -\frac{N\mu_o Ze^2}{6m}\langle r^2\rangle \tag{8.20}$$

This is the classical Langevin result and shows that the diamagnetic susceptibility is independent of temperature.

8.3.2 Quantum Theory

Using quantum mechanics, it can be shown that, in the presence of a magnetic induction **B**, the Hamiltonian $\mathcal{H}$ of an atomic electron changes to $\mathcal{H} + \mathcal{H}'$ where the term $\mathcal{H}'$ may be treated as a small perturbation and is given by

$$\mathcal{H}' = -\frac{ie\hbar}{2m}(\nabla.\mathbf{A} + 2\mathbf{A}.\nabla) + \frac{e^2}{.2m}A^2 \tag{8.21}$$

where A is the magnetic vector potential expressed by

$$\mathbf{B} = \nabla \times \mathbf{A}$$

If the magnetic induction is uniform, it can be described by the vector potential **A** as

$$\mathbf{A} = \frac{1}{2}\mathbf{B}\times\mathbf{r}$$

$$= \frac{1}{2}\left[\hat{\mathbf{i}}(B_y z - B_z y) + \hat{\mathbf{j}}(B_z x - B_x z) + \hat{\mathbf{k}}(B_x y - B_y x)\right] \tag{8.22}$$

For the magnetic field pointing along the z-direction, we have

$$B_x = B_y = 0,\ B_z = B,$$

and Eq. (8.22) gives

$$A_x = -\frac{1}{2}yB,\ A_y = \frac{1}{2}xB,\ A_z = 0 \tag{8.23}$$

Also,

$$\nabla.\mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} = 0$$

and

$$\mathbf{A}.\nabla = A_x\frac{\partial}{\partial x} + A_y\frac{\partial}{\partial y} + A_z\frac{\partial}{\partial z}$$

$$= \frac{1}{2}B\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right)$$

Therefore, Eq. (8.21) becomes

$$\mathcal{H}' = -\frac{ie\hbar B}{2m}\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right) + \frac{e^2B^2}{8m}\left(x^2 + y^2\right) \tag{8.24}$$

In quantum mechanics, the angular momentum operator is given by

$$\mathbf{L} = -i\hbar\,\mathbf{r} \times \nabla$$

$$= -i\hbar\left[\hat{\mathbf{i}}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right) + \hat{\mathbf{j}}\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right) + \hat{\mathbf{k}}\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right)\right] \tag{8.25}$$

Thus, it follows that the first term on the right hand side of (8.24) is proportional to L_z. The distance r is measured from the nucleus.

If we consider a unit volume of a substance containing N atoms each having Z electrons, the second term in the expression (8.24) becomes

$$\frac{Ne^2B^2}{8m}\sum_{i=1}^{Z}\left(x_i^2 + y_i^2\right) = \frac{NZe^2B^2}{8m} < \rho^2 > \tag{8.26}$$

where $<\rho^2>$ represents the mean of the squares of radii of the projections of orbits on a plane perpendicular to the magnetic field. Now if the magnetic field induces a dipole moment in the material, the corresponding energy term should be of the second power in B. Hence the term (8.26) may be regarded as the energy term associated with diamagnetism of the material. Comparing (8.26) with the diamagnetic energy, $\frac{1}{2}\mathbf{M}.\mathbf{B}$ or $-\frac{1}{2\mu_0}\chi_{dia}B^2$ we get,

$$\chi_{dia} = -\frac{N\mu_0 Ze^2}{4m} < \rho^2 > \tag{8.27}$$

Using Eq. (8.18), it becomes

$$\chi_{dia} = -\frac{N\mu_0 Ze^2}{6m} < r^2 > \tag{8.28}$$

where $<r^2>$ represents the mean square distance of the electrons relative to the nucleus. The result is identical to the one obtained from the classical theory. Thus, in order to calculate χ_{dia}, we must determine $<r^2>$ which